
PESKAS

Digital Monitoring System for Small Scale Fisheries

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Overview

Presentation topics

- Challenges in data collection and data management in small scale fisheries context
- **PESKAS**: a data management framework for small scale fisheries
 - Automated analytics framework data flow
 - The infrastructure
 - Showcase



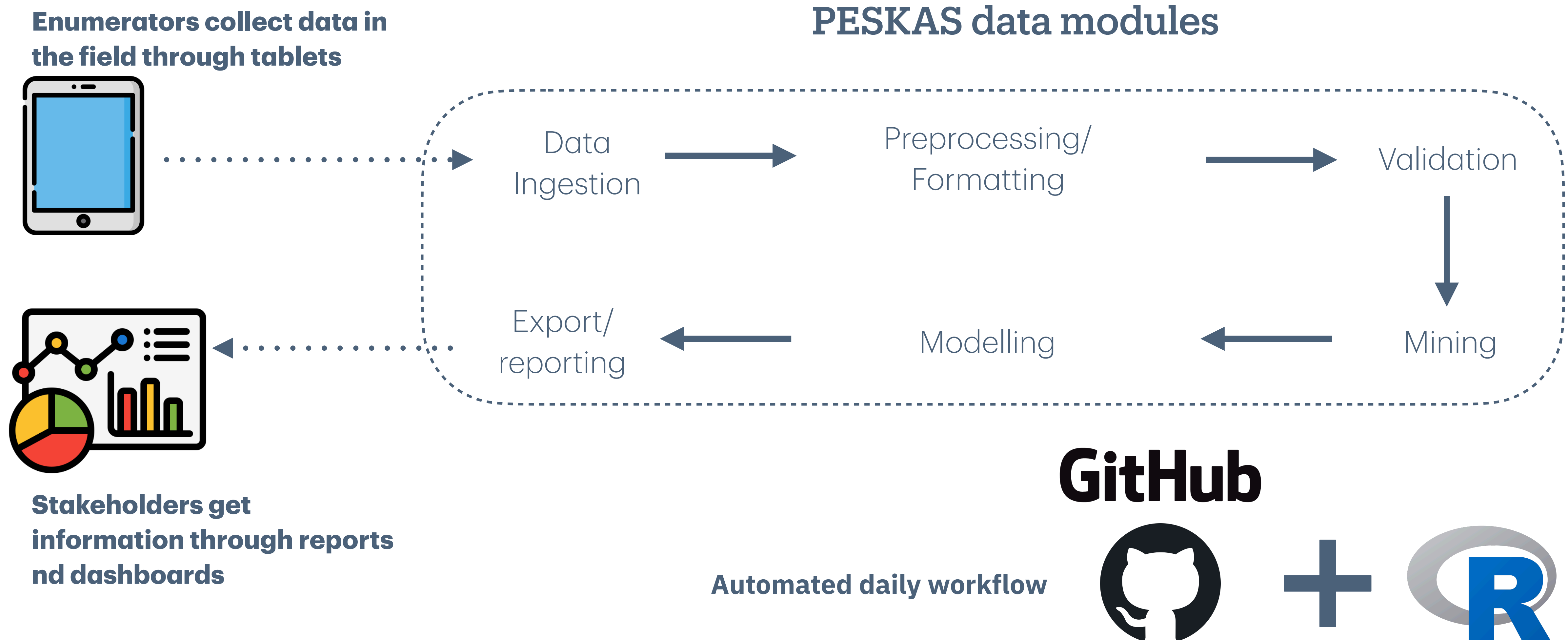
Challenges in SSFs data collection

Why Good Data Is Hard to Get

- **Paperbased Systems:** Physical forms vulnerable to damage, loss, and transcription errors
- **Inconsistent Methodologies:** Varying approaches to species identification and measurement
- **Data Verification Challenges:** Difficult to validate information from informal sector operations
- **Informal Nature:** Much activity occurs outside formal regulatory frameworks
- **High Diversity:** Varied gear types, target species, and practices require complex data structures

Automated Analytics Framework

PESKAS approach



The data

Key Variables collected

General trip info

Date of Collection

Landing Site / Area

Vessel / Boat Type

Gear Type

Number of fishers for fishing trip

Duration of fishing trip (hours)

Catch data

Fish group

Fish length (cm)

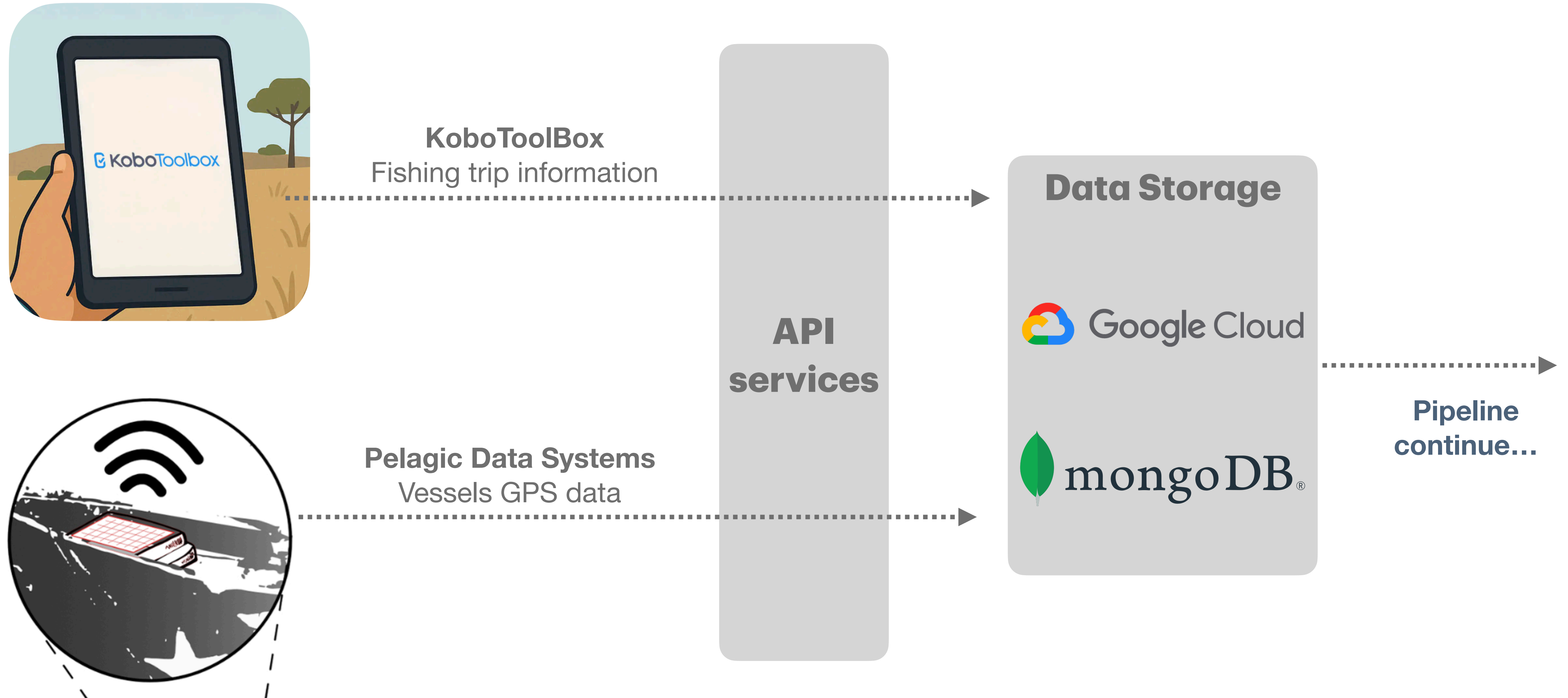
Catch price

Catch habitat



Data Ingestion

Getting data from multiple sources (running everyday)



Data Preprocessing

Make data usable and comparable

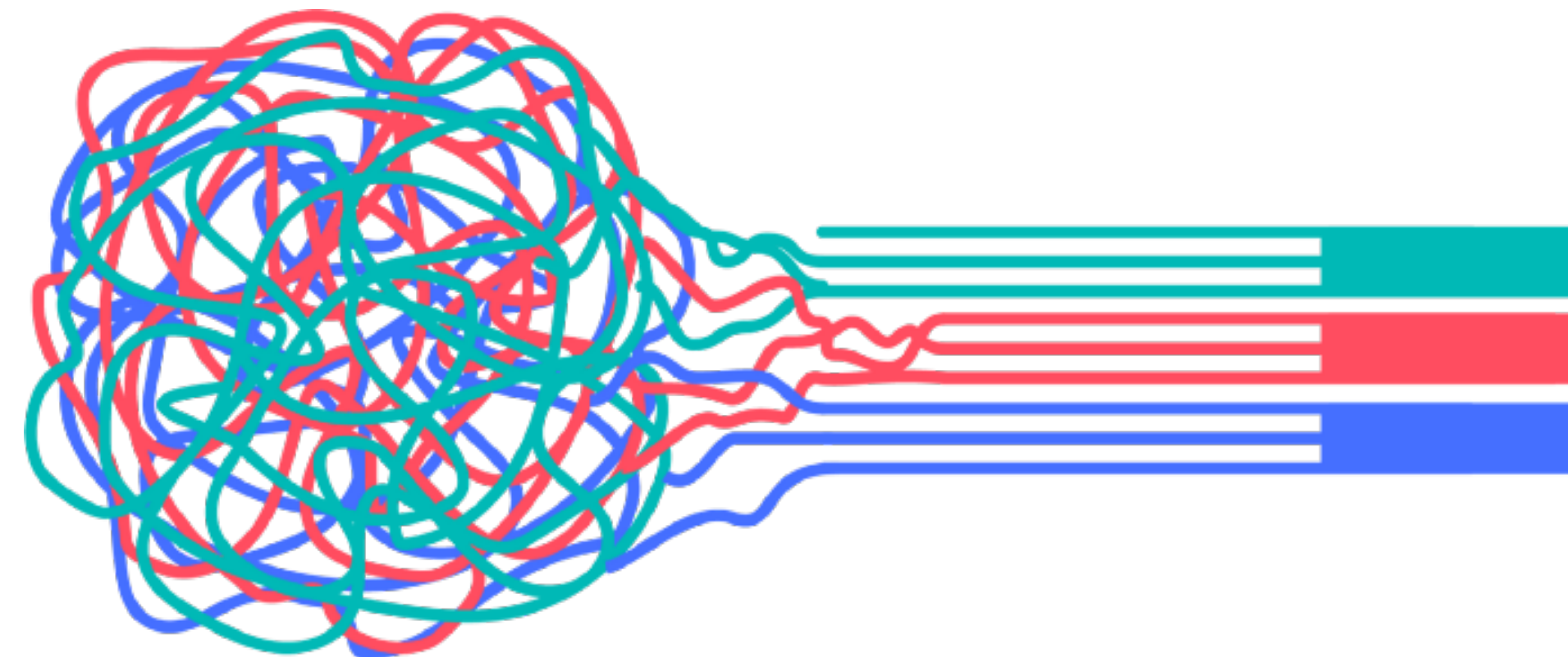
Data Cleaning & Formatting

- Removes redundant information and raw data artefacts
- Standardizes data formats (tidy format, apply recognised variable names conventions)
- Assigns appropriate fields to data (use ISO 8601 format for dates etc...)

Terminology Standardization

- Aligns with Aquatic Foods Ontology
- Consistent naming across variables
- Facilitates crossregion and crosscountry comparison

**Raw / Unformatted
data**



**Structured and
usable data**

Data Preprocessing

Ontology and semantics

Ensures each variable name has a single, clear definition and the various data collections speak the “same language”

Variables name: **Aquatic Food Ontology**

Variables values: **Agree on a shared coding system for each variable**

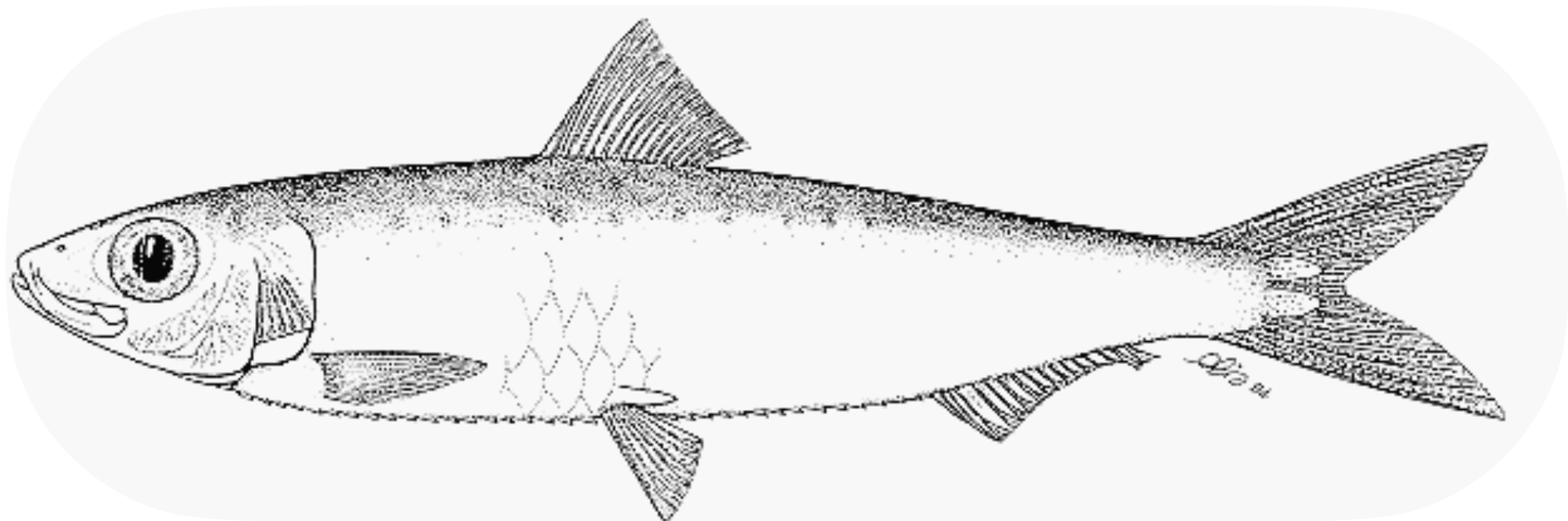
Gear Type → “LL” (long line), “GN” (gill net), etc.

Vessel Type → “CAN” (Canoe), “DHO (Dhow), etc.



Harmonising taxonomic information using FAO standards

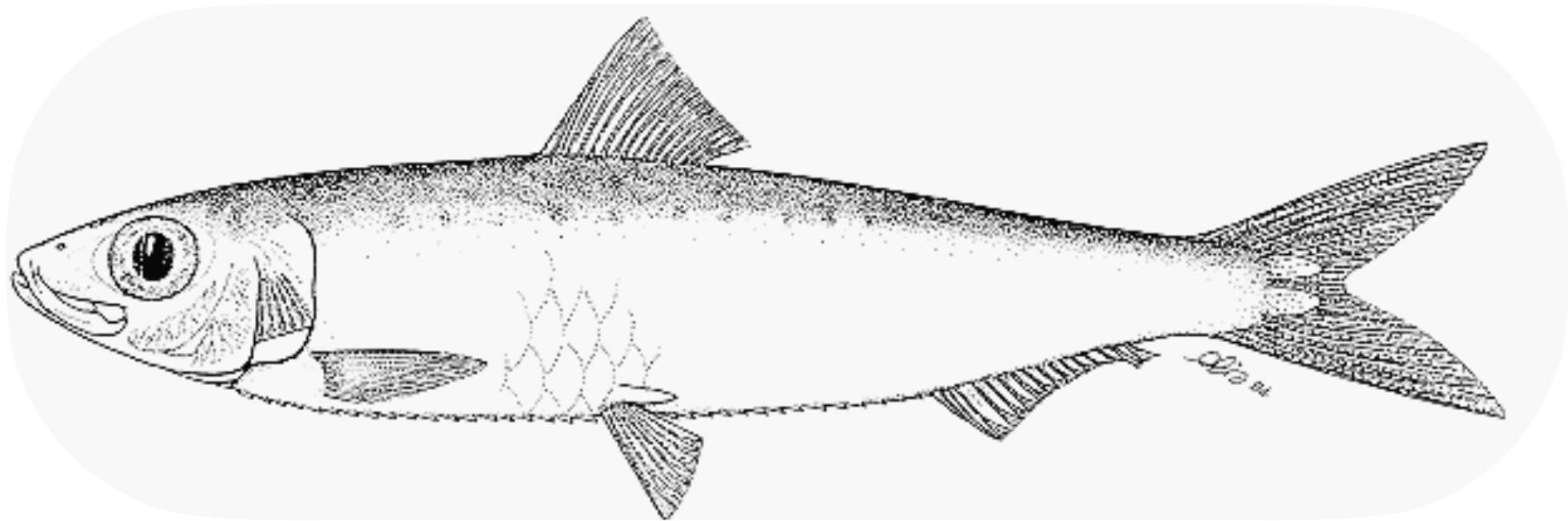
Sardina pilchardus



Hello, my name is:

Sardina
Sardine
Pilchard
Sardinopachilena
Sardella
Sardinha

Sardina pilchardus



Hello, my name is:

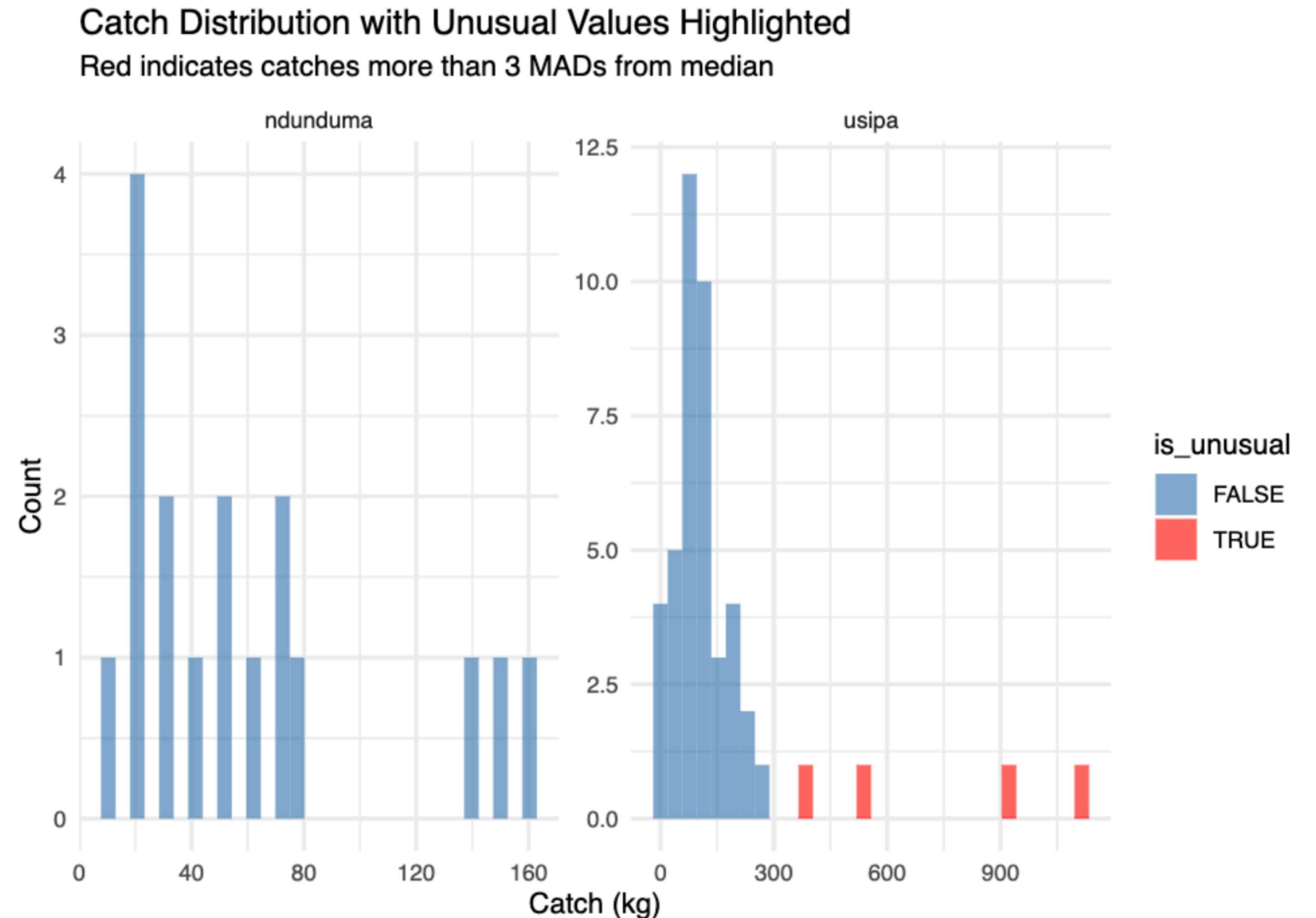
PIL
FAO 3alpha code

Data Validation

Automated Outlier Detection

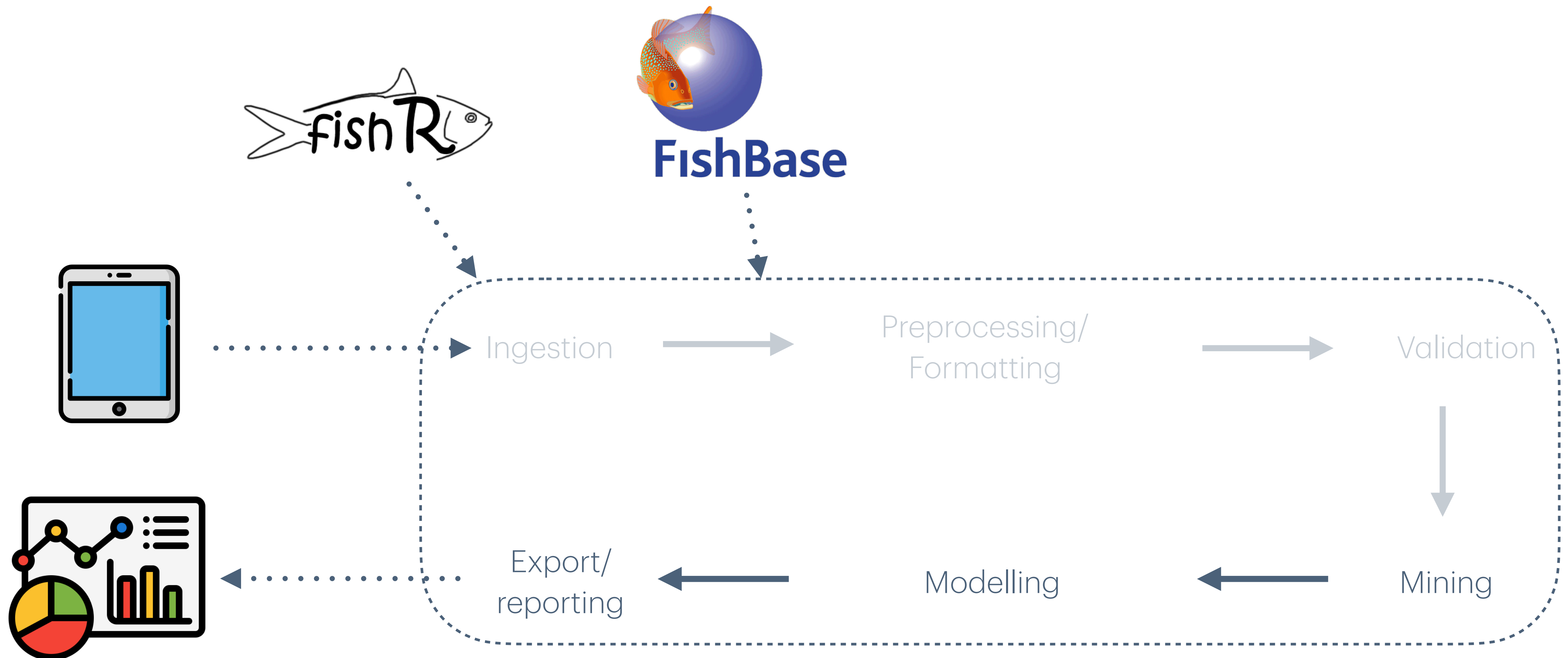
Data come with human errors:

- Overestimated/underestimated fish parameters
- Overestimated/underestimated revenue
- Anomalous fishing trip information
- Identifies anomalous values using statistical methods
- Employs both univariate (MAD) and multivariate approaches
- Automatic reporting of enumerators performance



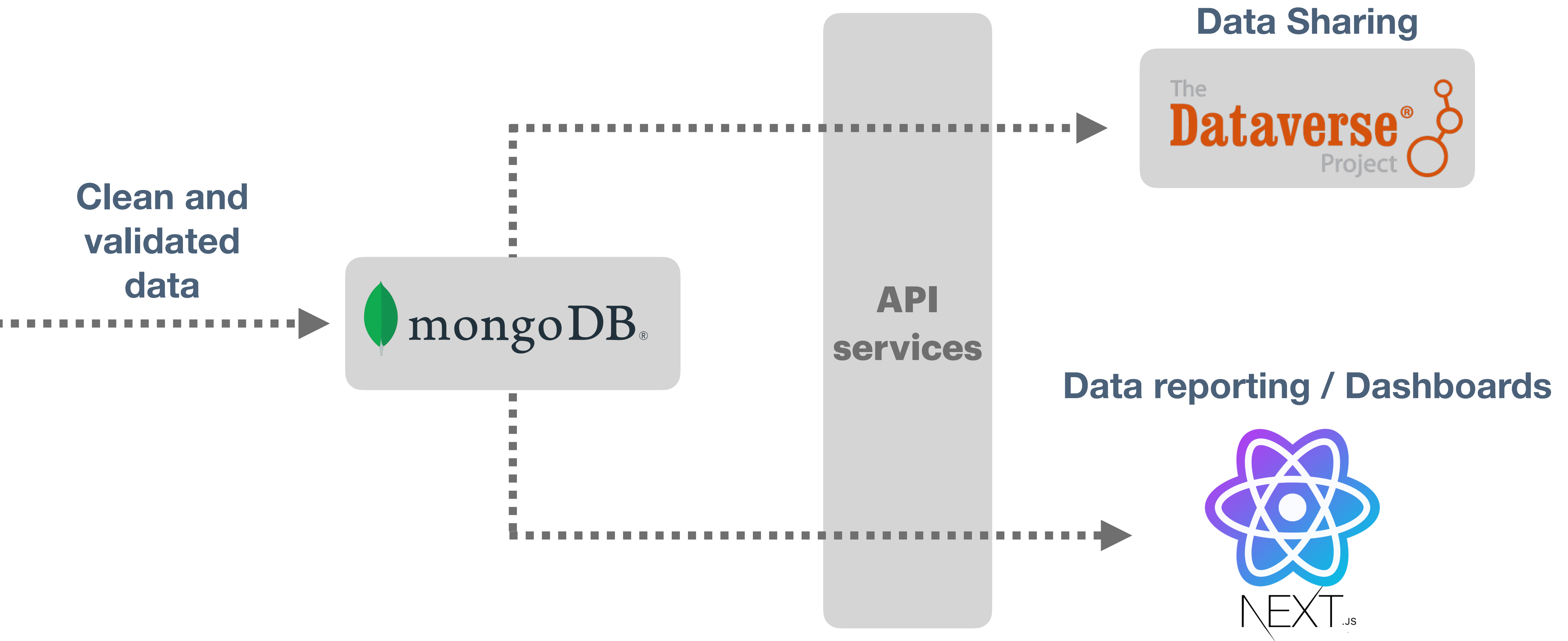
PESKAS approach

Data Mining and Modelling



Data Export

Sharing and Reporting



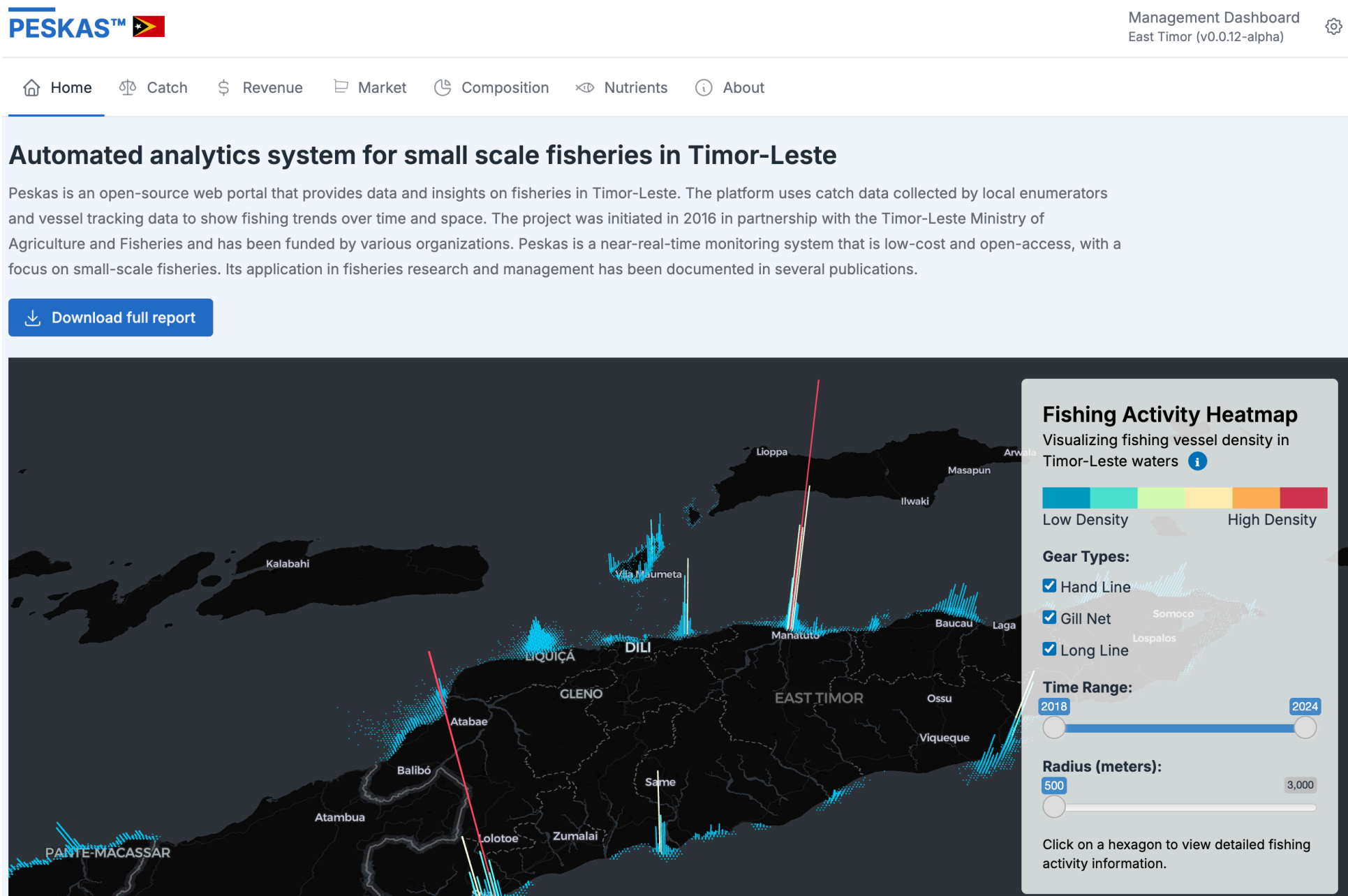
PESKAS Journey

From Timor-Leste (Asia) to East Africa

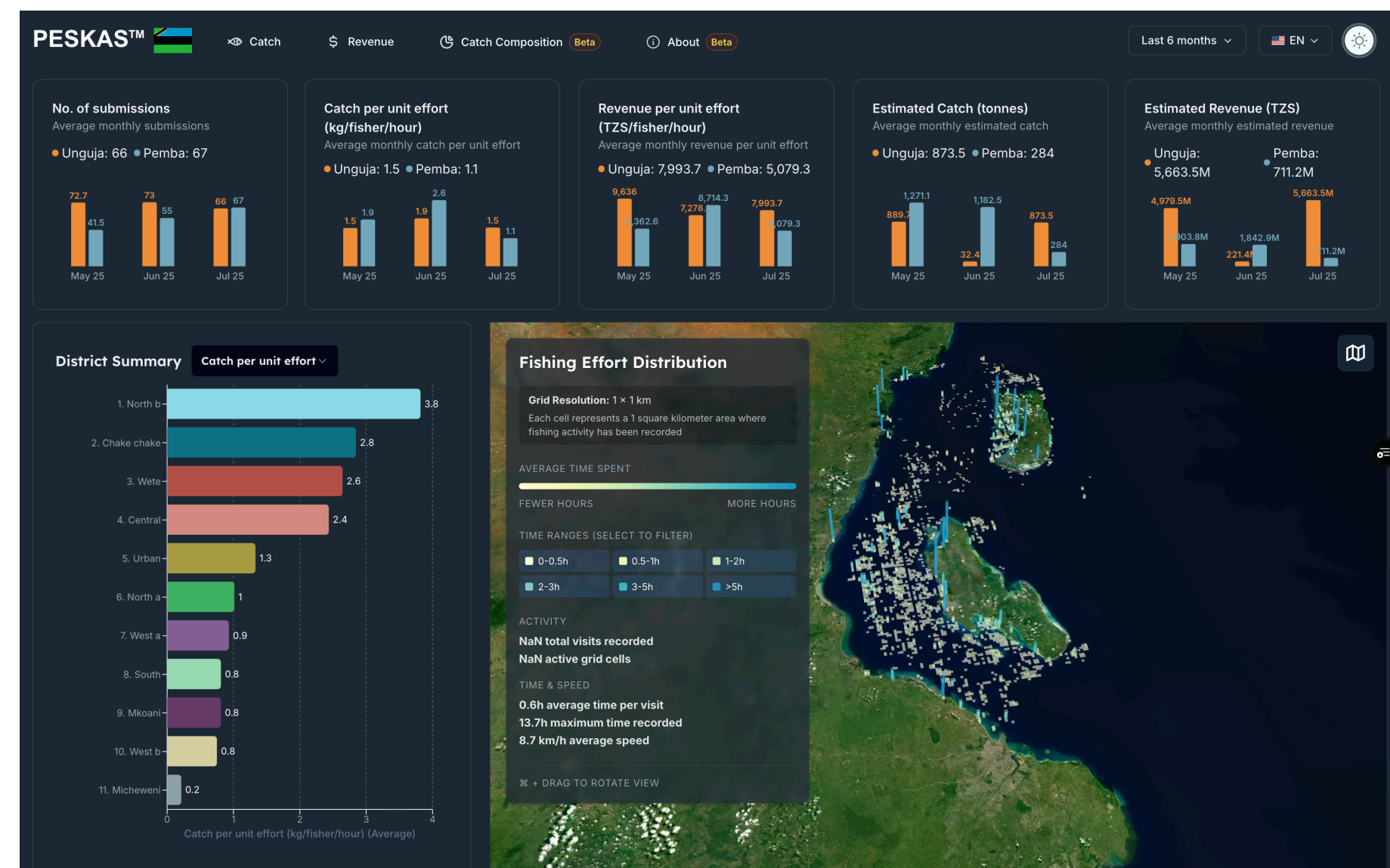


Some examples

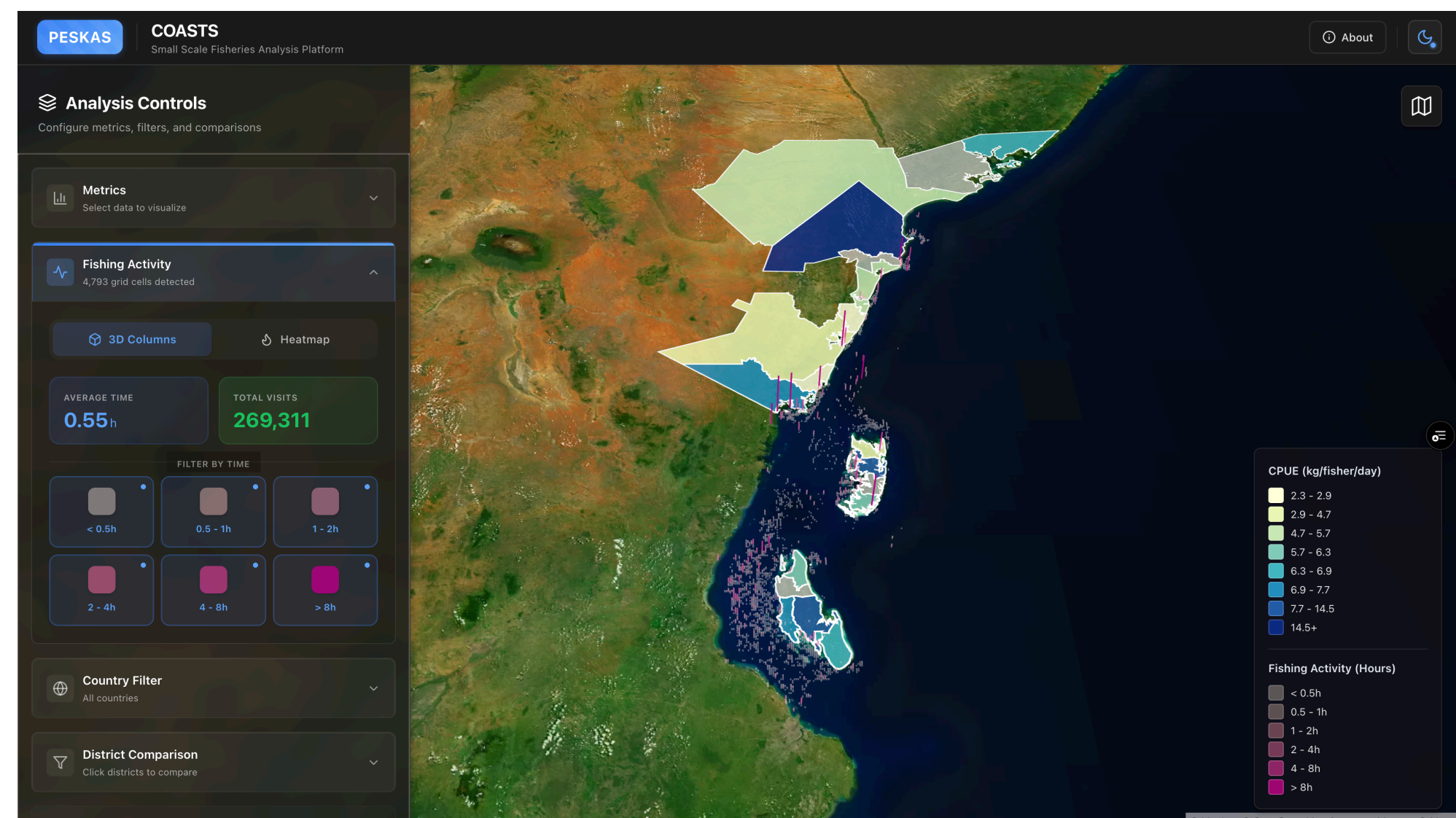
PESKAS Timor Leste



PESKAS Zanzibar



COASTS





↕
Data
storage



- TimorLeste
- Kenya
- Zanzibar
- Mozambique

GitHub

- Version Control
- Continuous integration
- Automated workflow with [GitHub Actions](#)

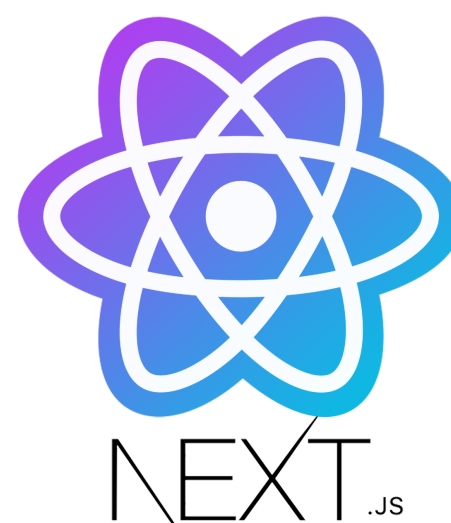
- R package
- Modular code



Community / Stakeholders
feedback



The
Dataverse[®]
Project



Data
sharing

Peskas web
portal application

Dynamic data
report

Thank you!